CENTRO ESTADUAL DE EDUCAÇÃO TECNOLÓGICA PAULA SOUZA

Jundiaí College of Technology

Higher Technology Course in Embedded Systems

THE PRESSURE GAUGE PROJECT (ETPMS)

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Jundiaí

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Project submitted to Jundiaí College of Technology as a requirement for passing the Integrative Project II course in the Higher Technology Degree in Embedded Systems program, under the supervision of Professor Ângela Lühmann.

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INRODUCTION

Pressure gauges are essential tools for drivers. Whether portable or stationary, they provide a means of checking whether a tire is inflated to the optimal pressure, or at least to a safe pressure, for use on motorized and non-motorized vehicles equipped with tires. With this in mind, the team responsible for this project set out to develop a pressure gauge that would be as comprehensive as, or more comprehensive than, commercially available models. The project builds upon existing designs for academic and educational purposes, given the valuable hands-on experience that its development would provide.

TEX Equipamentos Eletrônicos generously provided equipment for building the prototype, as well as in-person instruction held at the college facilities. The classes were taught by Henrique Coser, Software and Firmware Engineering Coordinator. Both forms of support helped reduce the development costs that would otherwise have been incurred.

Regarding the sensors, the prototype incorporates two pressure sensors designed for different environments. The first is the SMP3011, a sensor designed to measure gas pressure, while the second is the BMP280, which measures atmospheric pressure. The device also features an OLED display with a resolution of 128 × 64 pixels and an SSD1306 controller. Both sensors and the display controller use the I²C communication protocol, simplifying device programming. The firmware is written in C using the ESP-IDF framework, developed by Espressif.

1. SYSTEM DEFINITION (Software Engineering I)

1.1. SYSTEM SCOPE

The system must successfully read data from the hydrostatic pressure sensor, convert the readings into a pressure measurement unit that is understandable to the user, and display the converted data on the OLED screen.

1.2. SYSTEM REQUIREMENTS

Functional Requirements:

- (FR01) Measure Pressure: The system must measure tire pressure in real time.
- (FR02) Display Pressure: The system must clearly display the measured pressure on the OLED display, accompanied by informative labels describing the data shown.
- (FR03) Indicate When Recalibration Is Required: The system must display warning messages on the OLED screen when the pressure falls outside the safe range.
- (FR04) Audible Alerts: The system must emit audible alerts through a buzzer or another mechanism to notify the user of conditions or problems requiring attention, such as excessively high tire pressure, low battery, connection failures, errors, and other relevant issues.
- (FR05) Allow Measurement Unit Selection: The system's logical design must allow the user to switch between measurement units, providing convenient access to the unit that best suits their needs.
- (FR06) Continuous Updates: The system must update pressure readings at regular intervals.
- (FR07) User Interface: The system must allow user interaction through physical buttons.

Non-Functional Requirements:

- (NFR01) Prevent Duplicate Button Inputs: The system must accurately register button presses, avoiding missed inputs or multiple registrations for a single press.
- (NFR02) I²C Communication: Communication between the sensor and the microcontroller must use the I²C protocol.
- (NFR03) Use a Rechargeable Battery: The system must include a rechargeable battery so that it can be used away from external power sources.
- (NFR04) Operate Without Complex Configuration: The system must be simple to operate and provide consistent, stable performance. Throughout the sensor's operating period, regardless of the operator, the system must automatically read the pressure value without requiring manual intervention.
- (NFR05) Operate Without a Power Outlet: Portability must allow the user to measure tire pressure in situations away from conventional facilities, such as during trips, without having to rely on service stations equipped with pressure-measuring tools.
- (NFR06) Compact Design: The system must be compact enough to ensure that transportation and use are neither difficult nor inconvenient.
- (NFR07) Lightweight Design: The hardware must have a sufficiently low overall weight to avoid inconveniencing the user during operation.

- (NFR08) Ease of Use: The system must provide simple, lightweight, and intuitive operation.
- (NFR09) Pressure Sensor: The system must use a hydrostatic pressure sensor.

2. SENSOR SPECIFICATION (Digital Electronics II)

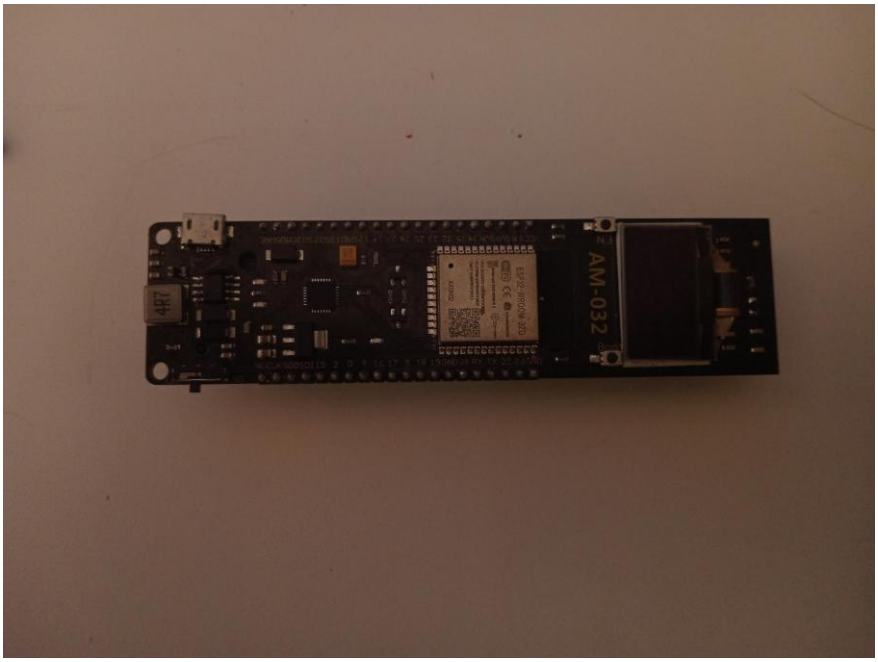
ESP32 Development Board with Integrated SSD1306 OLED Display (ANTS WAKE AM-032)

The ANTS WAKE AM-032 development board is based on the ESP32 microcontroller, a high-performance device capable of running embedded applications with built-in Wi-Fi and Bluetooth connectivity.

In addition to the microcontroller, the board features an integrated SSD1306 OLED display, allowing information to be displayed directly on the hardware without requiring external display modules. Communication between the ESP32 and the display is handled through the board's internal I²C interface.

The ESP32 operates at a logic voltage of 3.3 V and provides several input/output capabilities, including I²C, SPI, UART, PWM, and analog-to-digital converters (ADCs), making it suitable for monitoring, automation, and Internet of Things (IoT) applications.

Figure 1 - ANTS WAKE AM-032



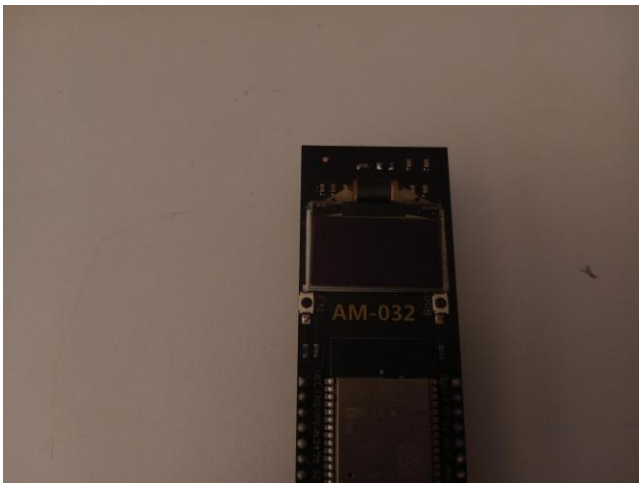
Source: Author's own work (2026)

SSD1306 Display

The integrated SSD1306 OLED display has a resolution of 128 × 64 pixels, low power consumption, and the ability to display text, numbers, graphics, and icons, facilitating the visualization of data collected by the sensors connected to the system.

The display is integrated directly into the ESP32 development board used in this project. It features a two-color OLED screen and communicates through the I²C protocol, as do the sensors described above. Its resolution is 128 × 64 pixels, and it operates at 3.3 V, the typical logic voltage of the ESP32.

Figure 2 – OLED SSD1306 Display

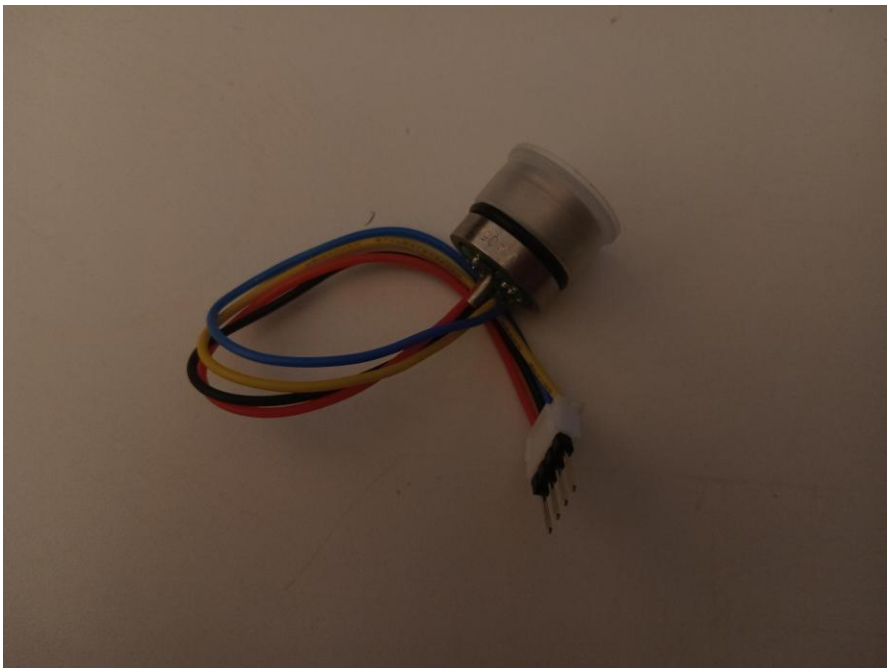


Source: Author's own work (2026)

SMP3011 Sensor

The SMP3011 is a sensor capable of measuring gas pressure and transmitting the corresponding reading to the system firmware via the I²C communication protocol. It uses the default I²C address 0x78 to communicate with the ESP32 microcontroller.

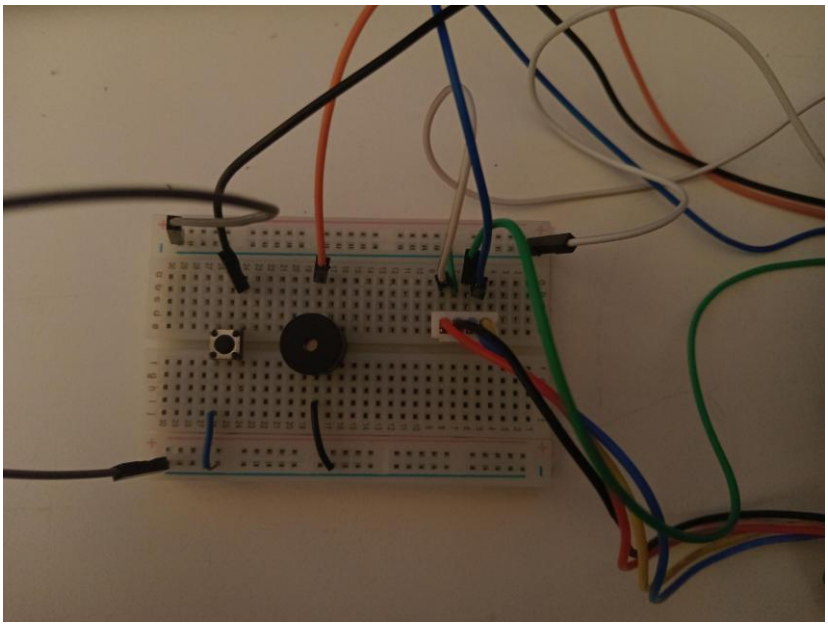
Figure 3 – SMP301



Source: Author's own work (2026)

3. SENSOR ASSEMBLY (Home Automation and Embedded Systems Programming I)

Figure 4 – Breadboard Wiring



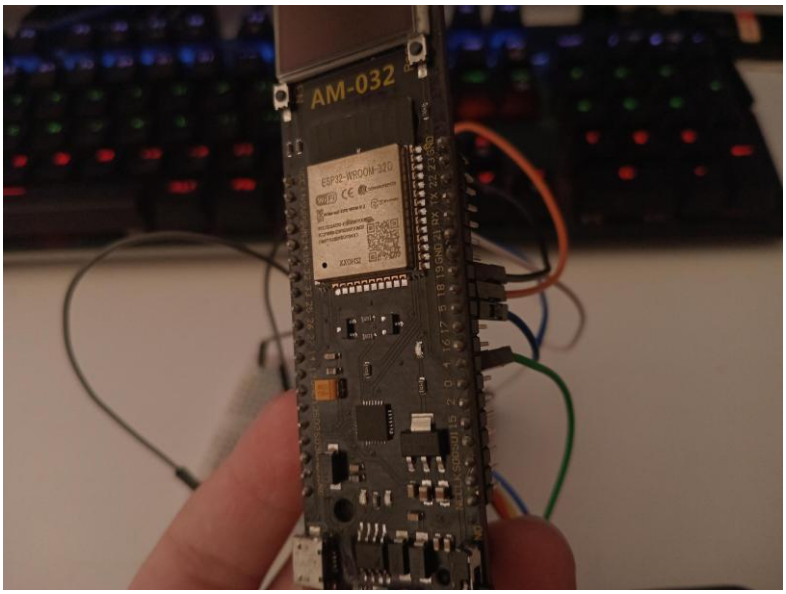
Source: Author's own work (2026)

The figure above shows a push button, a piezo buzzer, and the connector used to connect the SMP3011 sensor. The push button is configured to use a digital input with the ESP32's internal pull-up resistor enabled. One of its terminals is connected to the microcontroller's input pin, while the other is connected to GND. This causes the input signal to be driven to a logic-low level when the button is pressed.

The piezo buzzer is used to emit audible warning signals. In simple applications, it can be connected directly to an ESP32 digital pin without additional resistors, provided that the buzzer's electrical requirements are compatible with the GPIO output. Additional components may be required for filtering or specific audio signal control.

The SMP3011 sensor uses four main connections for communication and power supply. The VCC terminal receives a 3.3 V supply voltage and is represented by the red wire. The GND terminal is connected to the circuit ground and is represented by the black wire. The SDA terminal, represented by the blue wire, handles data transmission and reception over the I²C protocol. The SCL terminal, represented by the yellow wire, receives the clock signal generated by the ESP32, synchronizing communication between the devices connected to the bus.

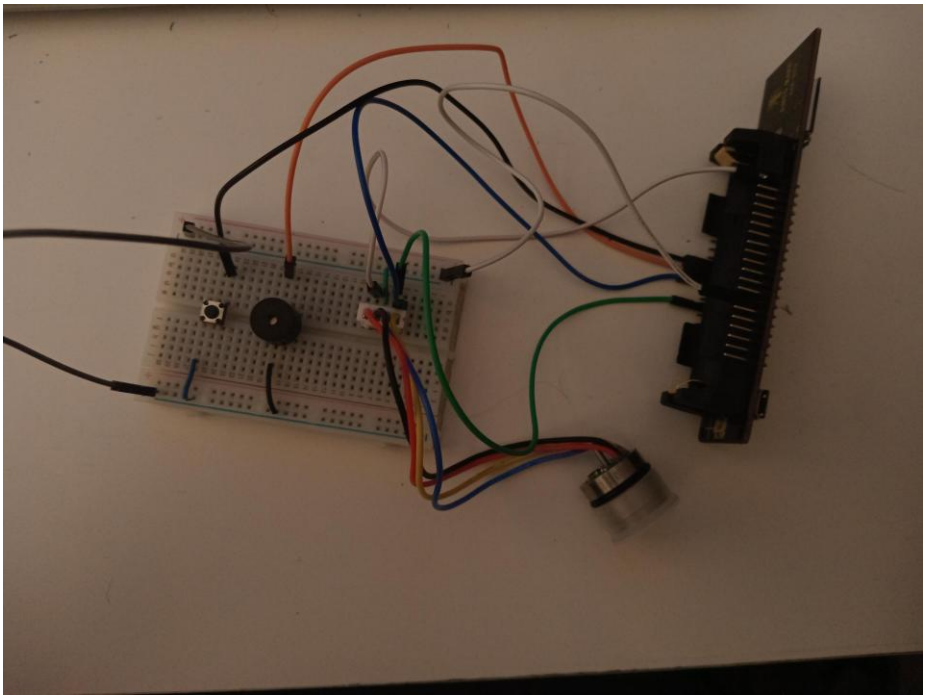
Figure 5 - ESP32 Ports



Source: Author's own work (2026)

As shown in the figures above, in addition to the VCC and GND power connections, the following connections were established between the devices and the ESP32: the push button was connected to GPIO 19, the piezo buzzer to GPIO 18, and the SDA line of the I²C bus to the GPIO pin.

Figure 6 – Pressure Sensor Prototype



Source: Author's own work (2026)

C Source Code

The firmware source code was developed in C using the ESP-IDF framework. To ensure maintainability and follow software engineering best practices, the architecture was modularized into independent components for OLED display management (LVGL), sensor readings via I²C, and hardware interrupts. The complete, documented, and updated source code is available in the source files hosted in the project's official repository.

CONCLUSION

The development of the tire pressure gauge prototype satisfactorily achieved the objectives established for the Integrative Project II course. By integrating the ESP32 microcontroller with the SMP3011 pressure sensor and the SSD1306 OLED display, the team was able to build a functional device capable of measuring pressure in real time.

The implementation of the firmware in C using the ESP-IDF framework ensured compliance with the previously established system requirements. The use of the I²C communication protocol enabled efficient data exchange between the hardware components. Furthermore, the logical architecture allowed direct user interaction through physical buttons, enabling the selection between measurement units (PSI and bar), as well as the precise activation of visual warnings on the display and audible alerts through the piezo buzzer whenever the pressure required adjustment.

In conclusion, the project not only validated the feasibility of building a compact, portable measuring device but also provided the team with valuable academic and hands-on experience. The practical application of concepts from Software Engineering, Digital Electronics, and Home Automation, combined with the essential support and equipment provided by TEX Equipamentos Eletrônicos, resulted in a stable and comprehensive embedded system that can serve as a foundation for future improvements.

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